Supervyse OPF Redfish Deep Operation User Guide

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Revision History

Revision Number	Description	Author	Reviewer	Release Date
1.0	Initial release.	Terry Hsu	Victor Wu	2024-12-10

1 Introduction

This document aims to assist developers, users, and testers in understanding the specifications and usage of Redfish Deep Operation.

1.1 Overview

The purpose of this document is to introduce in detail the design specifications of Redfish Deep Operation and how to correctly use these operations for system management.

1.2 Related Information

The following publications and sources of information are referred to by this document or may be useful to you:

- *Redfish Specification DSP0266, DMTF*

2 Deep Operation

2.1 Description

Implementations may support operations that modify the current resource as well as subordinate resources.

These operations are known as deep operations.

They give the client the ability to modify more than one resource with a single operation.

- Services that support deep PATCH for updating resources shall set the value of the DeepPATCH property in the DeepOperations property in the ProtocolFeaturesSupported property within the service root to `true`.
- Services that support deep POST for creating resources shall set the value of the DeepPOST property in the DeepOperations property in the ProtocolFeaturesSupported property within the service root to `true`.
- If the service supports deep operations, the MaxLevels property in the DeepOperations property in the ProtocolFeaturesSupported property in the service root shall indicate the maximum number of levels that the service supports for deep operations.
- To request deep operations on a resource, send the HTTP method to the deep operation URI of the resource. The URI for deep operations on any resource shall be in the format: `<ResourceUri>.Deep`.
- Requests that include references to multiple instances, such as members of a collection, shall include the `Members` property as part of the request body.

3 Deep POST

3.1 Description

Create one or more resources in a resource collection and optionally subordinate resources.

Except for the following two points:

- Action URIs shall not support deep POST operations.
- Deep POST shall not be allowed on the SessionCollection resource.

3.2 Example

The following deep POST example creates four members in the InsydeIPAccessCollection resource.

```
curl -sku root:0penBmc -X POST \
https://127.0.0.1/redfish/v1/Managers/bmc/Oem/Insyde/IPv4AccessCollection.Deep -d \
{ \
   "Members": [ \
      { \
         "Type": "IPMASK", \
         "Policy": "ACCEPT", \
         "IPMask": { \
            "IP": "192.168.1.1" \
         } \
      }, \
      { \
         "Type": "IPRANGE", \
         "Policy": "DROP", \
         "IPRange": { \
            "Start": "192.168.1.1", \
            "End": "192.168.1.2", \
         } \
      }, \
      { \
         "Type": "MAC", \
         "Policy": "DROP", \
         "MAC": "AA:BB:CC:DD:EE:FF" \
      }, \
      { \
         "Type": "PORT", \
         "Policy": "DROP", \
         "Port": { \
            "Protocol": "TCP", \
            "Start": 1000, \
            "End": 1000 \
         } \
      } \
   ] \
}
```

4 Deep PATCH

4.1 Description

Modify a resource and one or more subordinate resources.

To determine which members of subordinate resource collections are to be modified by a deep PATCH, services shall use the `@odata.id` property provided by the client to identify the member of the resource collection to be modified.

4.2 Example

4.2.1 For Collection

The following deep PATCH example modifies two members of the InsydeIPAccessCollection resource.

```
curl -sku root:0penBmc -X PATCH -H 'Content-Type: application/json' \
https://127.0.0.1/redfish/v1/Managers/bmc/Oem/Insyde/IPv4AccessCollection.Deep -d \
{ \
   "Members": [ \
      { \
         "@odata.id":"/redfish/v1/Managers/bmc/Oem/Insyde/IPv4AccessCollection/2", \
         "Type": "IPMASK", \
         "Policy": "ACCEPT", \
         "IPMask": { \
            "IP": "192.168.1.1" \
         } \
      }, \
      { \
         "@odata.id":"/redfish/v1/Managers/bmc/Oem/Insyde/IPv4AccessCollection/1", \
         "Type": "IPRANGE", \
         "Policy": "DROP", \
         "IPRange": { \
            "Start": "192.168.1.1", \
            "End": "192.168.1.2", \
         } \
      } \
   ] \
}
```

4.2.2 For Others

The following deep PATCH example modifies two members of the ServiceRoot resource

```
curl -sku root:0penBmc -X PATCH -H 'Content-Type: application/json' \
https://127.0.0.1/redfish/v1.Deep -d \
{ \
    "AccountService": { \
        "@odata.id": "/redfish/v1/AccountService", \
      "AccountLockoutThreshold": 4, \
      "AccountLockoutDuration": 30 \
   } \
   "SessionService": { \
```

```
        "@odata.id": "/redfish/v1/SessionService", \
        "SessionTimeout": 30 \
        } \
}
```

5 Reference

Redfish Specification DSP0266, DMTF #7.13